Build Your Own AI Career Coach

A hands-on cloud and AI workshop · 45 minutes



Today, you don't just watch. You build.



Build a real AI app

You will connect cloud tools to make an app that works.



Put it online

Your app gets its own web link that anyone can open.



Get your own plan

The app makes a step-by-step plan for your tech career.

What is the cloud?

The cloud is computers you use over the internet.

- Companies like Amazon own huge buildings full of computers. These are called data centers.
- You can use their computers from your own laptop or phone.
- You only pay for what you use, like a phone plan.



Think of it like electricity

You don't build a power plant at home. You plug in and pay for what you use. The cloud works the same way for computers.

You already use the cloud every day

Streaming

Shows and music play from the cloud, not your phone.

Photo backup

Your pictures are saved online so you don't lose them.

Email

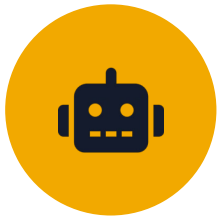
Your messages live online. You can read them anywhere.

Group chats

Messages travel through the cloud to your friends.

If it works on any device after you log in, it probably lives in the cloud.

What is AI?



AI (artificial intelligence) is software that learns from lots of examples.

Generative AI makes new things

It can write, draw, and plan. You ask in plain words. It answers.

We don't build the AI "brain"

That takes huge amounts of data and money. Big companies have already done it.

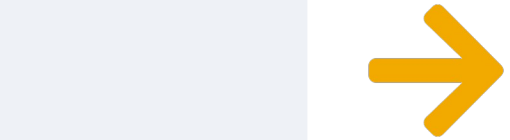
We connect to it

Our app sends the AI a question and shows you its answer. That's a real skill companies need.

Our project: AI STEM Career Navigator

You tell it

- The tech job you want
- Your experience so far
- What you're interested in
- How many hours a week you can study



AI gives you

- Skills to learn, in order
- Certifications to earn
- Projects to build
- A 90-day plan made for you

Why we built this

"Where do I even start?"

Many of us don't have someone in tech we can ask.



Tech is a big world

It's easy to feel lost. There are so many jobs, classes, and certificates.



A clear first step

This app gives you a starting point made just for you.



Create, don't just consume

Building it proves you can make things with AI, not just use them.

What we hope you take home



A working app

Your own live link to share with anyone.



Real skills

How cloud tools connect to make an AI app.



Your roadmap

A plan for your next 90 days in tech.



Confidence

Proof that you can build with AI. Yes, you.

Meet the 4 AWS tools we'll use



AWS Amplify

Puts our website on the internet so anyone can visit it.



API Gateway

The front door. It takes requests from the website and passes them on.



AWS Lambda

Runs our Python code, only when someone clicks the button.



Amazon Bedrock

Lets us use a smart AI model, called Amazon Nova, to write the plan.

No servers to buy. No AI to train. We just connect the pieces.

AWS Amplify: a home for our website



Amplify puts websites on the internet.

NEW WORD

Hosting: keeping a website online so people can visit it.



What it does in our app

It holds our 3 website files and gives us a web link that anyone can open.



Think of it like

Renting a storefront on a busy street. People can walk in any time, day or night.



Why it helps

You just upload a zip file. Amplify keeps the site fast and secure for you.

API Gateway: the front door



API Gateway lets websites and apps talk to our code.

NEW WORD

API: a way for two computer programs to talk to each other.



What it does in our app

When you click Generate, the website sends your answers to one web address. API Gateway receives them and passes them to Lambda.



Think of it like

The front desk of a building. It greets visitors and sends them to the right room.



Why it helps

It keeps our code safe behind one door. It can also limit how many requests come in at once.

AWS Lambda: our code, ready when needed

Lambda runs code without you owning a computer.

NEW WORD

Serverless: you don't manage any servers. AWS does it for you.

What it does in our app

Our Python code reads your answers, writes clear instructions for the AI, and sends the AI's answer back to you.

Think of it like

A cook who only comes in when an order arrives. No orders means no cost.

Why it helps

No servers to buy or fix. You only pay for the time your code is running.

Amazon Bedrock: the AI brain



Bedrock lets apps use powerful AI models.

NEW WORD

Prompt: the instructions you give an AI.



What it does in our app

It sends our instructions to an AI model called Amazon Nova Pro. Nova writes your skills, certifications, projects, and 90-day plan.



Think of it like

A team of expert chefs. You pick one, tell them what you want, and they create it.



Why it helps

We don't have to train our own AI, which costs millions. Each answer costs a tiny amount.

Two helpers working behind the scenes



AWS IAM

Decides who is allowed to do what.

We give Lambda a permission that says: you may use the Nova AI. Nothing else.

Think of it like: *A key card that only opens certain doors.*



Amazon CloudWatch

Keeps a record of what our code did.

If something breaks, we read these records, called logs, to find out why.

Think of it like: *A notebook that writes down everything that happens.*

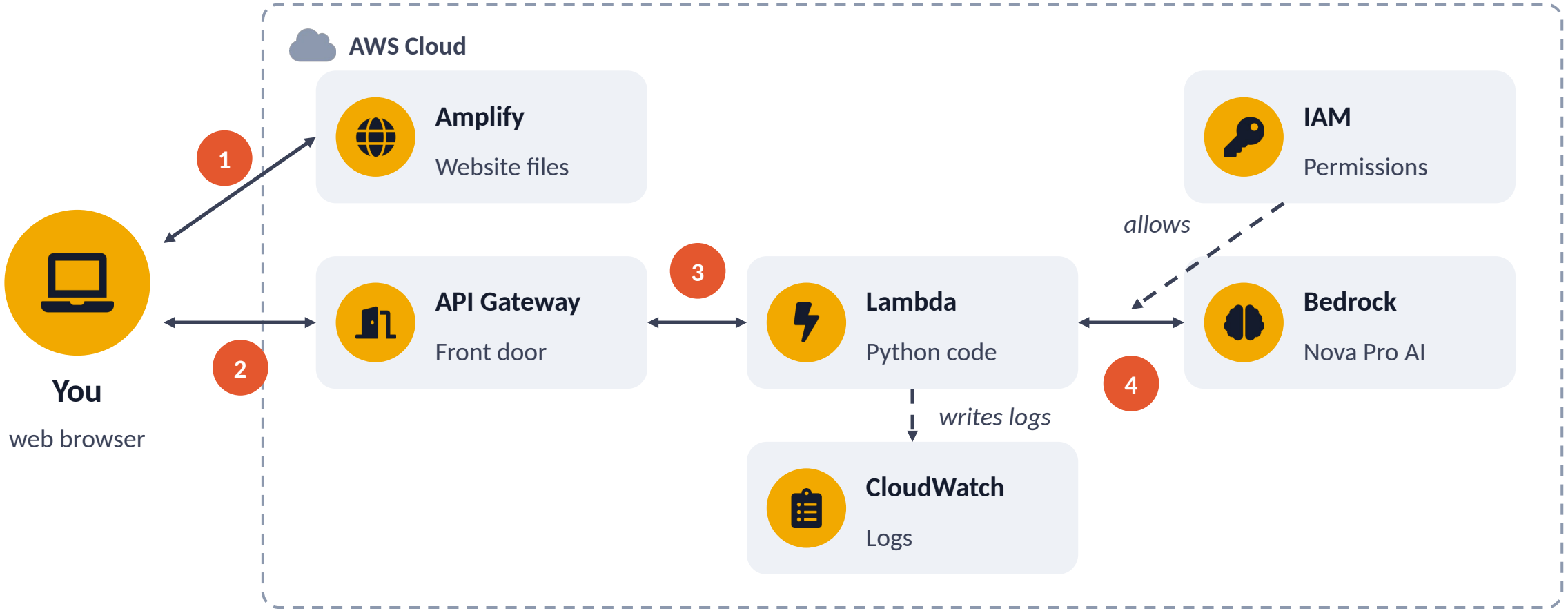
Putting it together: it's like a restaurant



You	Amplify	API Gateway	Lambda	Bedrock
Order	**The dining room**	**The server**	**The cook**	**The chef**
You fill out the form and click Generate.	The website where you place your order.	Takes your order to the kitchen.	Gets the order ready for the chef.	The AI creates your plan.

Then your meal comes back the same way: your roadmap shows up on screen in about 5 seconds.

The real map: how the pieces connect



1. Open the website
2. Send your answers
3. Lambda writes the prompt
4. Nova writes your roadmap
5. Roadmap comes back to you

Our plan for the next 45 minutes

1	0-5 min	Learn how cloud AI apps work
2	5-10 min	Try the AI in Amazon Bedrock
3	10-20 min	Build the Lambda function
4	20-28 min	Connect API Gateway
5	28-37 min	Connect the website
6	37-41 min	Test your app
7	41-44 min	Put it online with Amplify
8	44-45 min	Wrap up and celebrate

You don't need to be an expert



What you need

A laptop, an AWS account, and your curiosity. That's it.



Errors are clues

Red error messages are normal. They tell us what to fix. They don't mean you failed.



Help each other

Stuck? Raise your hand. Finished early? Help your neighbor.

Where these skills can take you

- Cloud Engineer
- AI Engineer
- Cybersecurity Analyst
- Data Analyst
- Solutions Architect
- Software Developer

Today you use the same kinds of tools people in these jobs use.

Let's build.

Get the files: github.com/kingmellow08/ai-career-navigator

